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1 Seebeck Effect

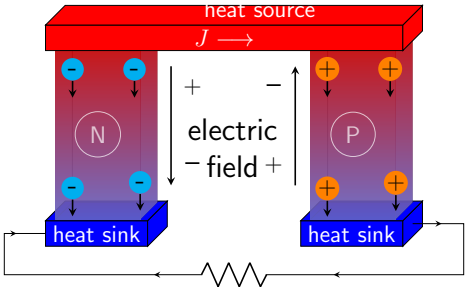

Figure 1: Seebeck effect showing thermoelectric voltage generation in N-type and P-type legs under a temperature gradient.

2 Peltier Effect

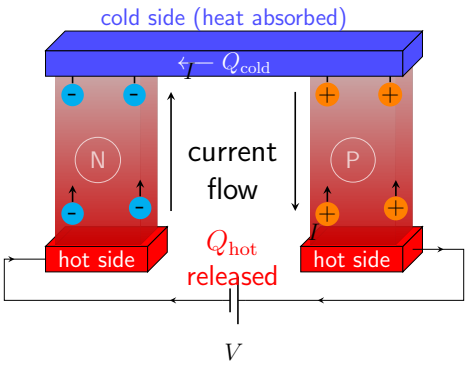

Figure 2: Peltier effect demonstrating heat absorption and release due to applied current in thermoelectric materials.

3 Landauer Transport Formalism

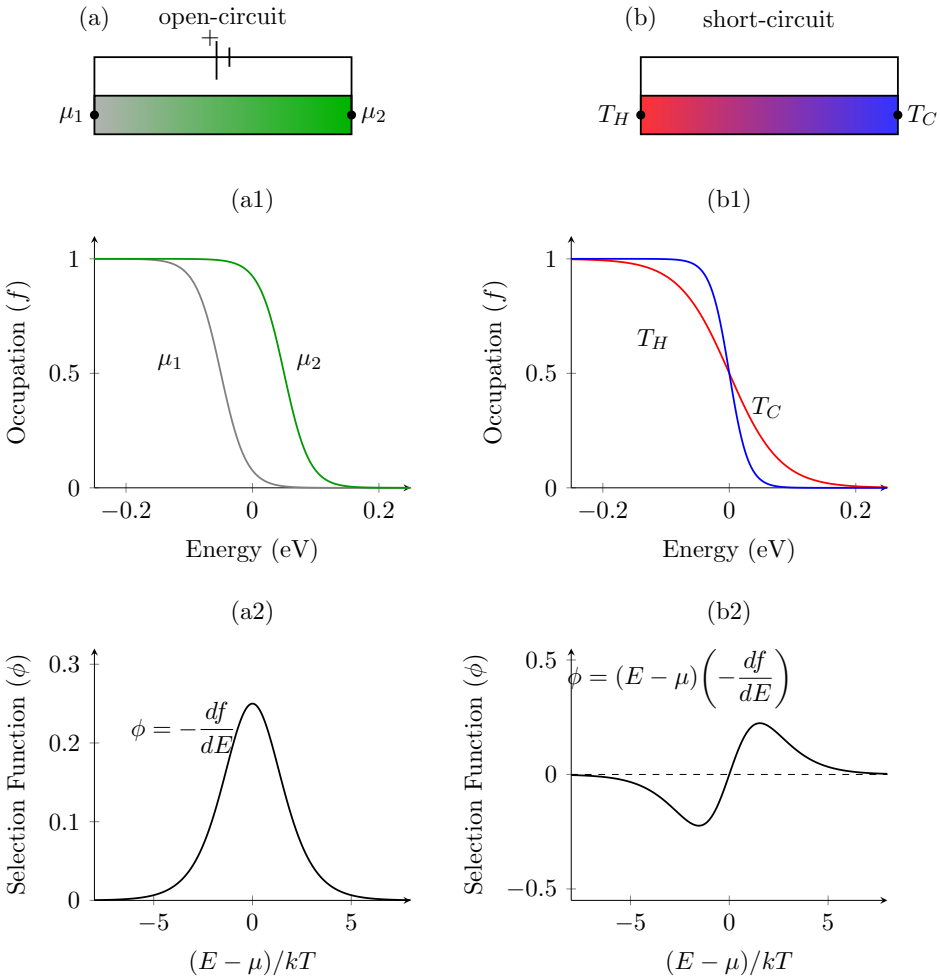

Figure 3: Landauer transport model illustrating open-circuit and short-circuit conditions along with energy-dependent selection functions.

4 Thermoelectric Figure of Merit (zT)

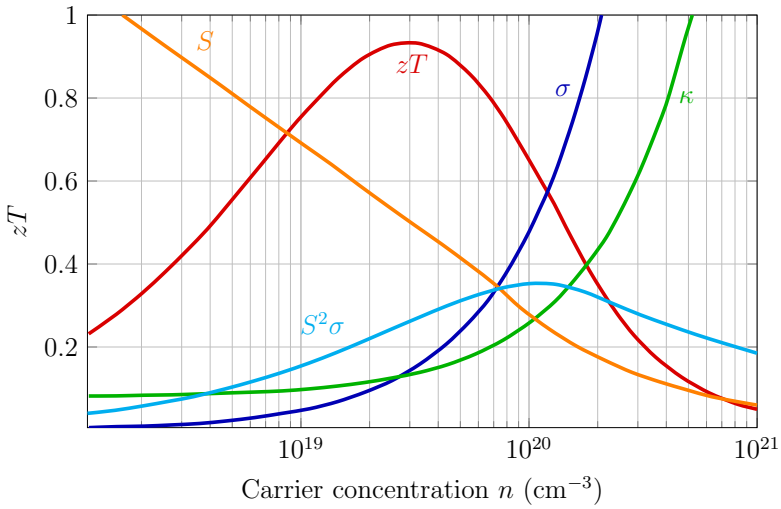

Figure 4: Variation of thermoelectric figure of merit (zT), electrical conductivity, thermal conductivity, and Seebeck coefficient with carrier concentration.

5 Degrees of Freedom in Thermoelectrics

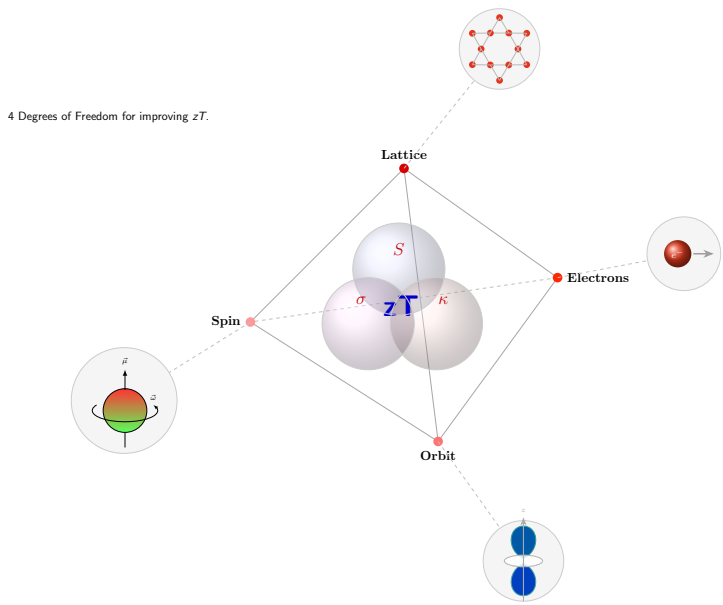

Figure 5: Interdependence of Seebeck coefficient, electrical conductivity, and thermal conductivity in determining thermoelectric performance (zT).

6 Point Defect Scattering

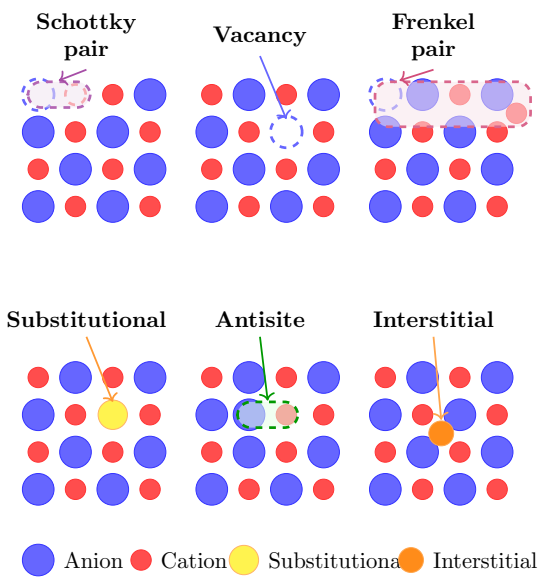

Figure 6: Different types of point defects including Schottky pairs, Frenkel pairs, vacancies, substitutional, antisite, and interstitial defects.

7 Thermoelectric Module Architecture

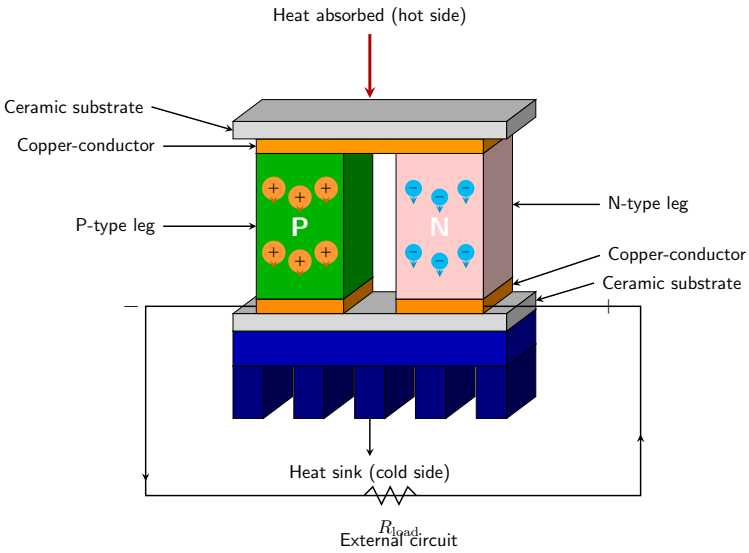

Figure 7: Three-dimensional schematic of a thermoelectric generator module showing P-type and N-type legs, electrodes, and heat flow.

8 Efficiency vs Heat Source Temperature

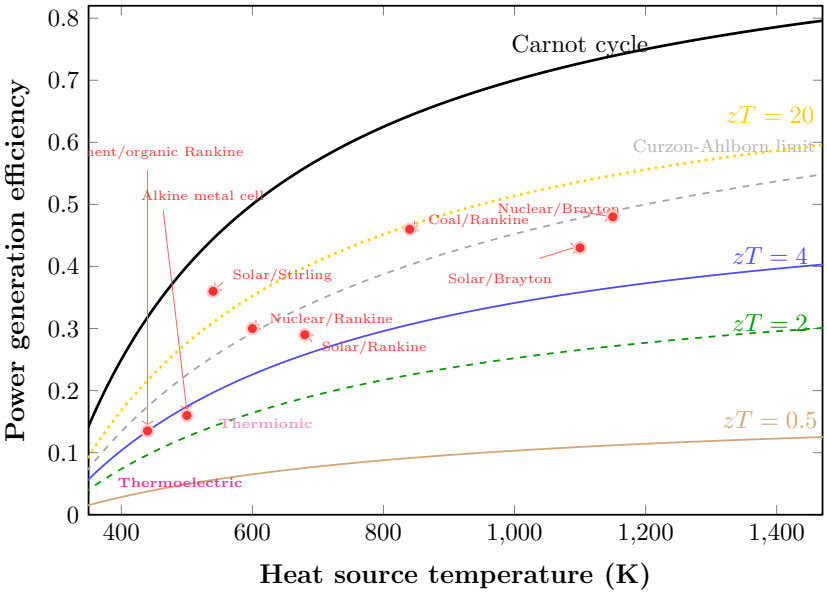

Figure 8: Efficiency comparison of thermoelectric devices with Carnot and Curzon–Ahlborn limits for different values of zT.

9 Laser Flash Analysis (LFA)

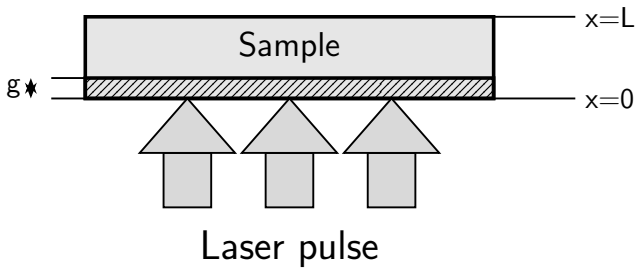

Figure 9: Laser flash analysis setup illustrating sample geometry and transient heat pulse propagation.

10 Parker Function

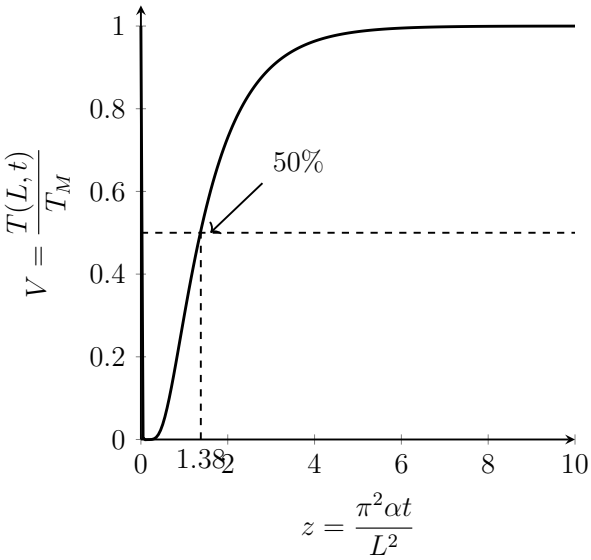

Figure 10: Normalized temperature rise curve (Parker function) used to determine thermal diffusivity.

11 UV-Vis-NIR Spectrometer

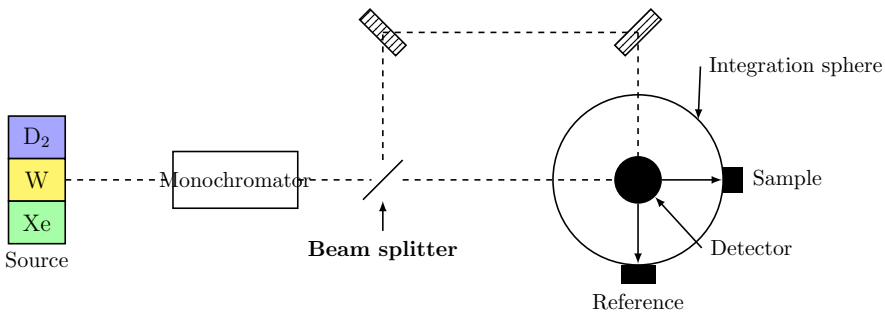

Figure 11: Schematic of a UV-Vis-NIR spectrometer with integrating sphere and detector configuration.